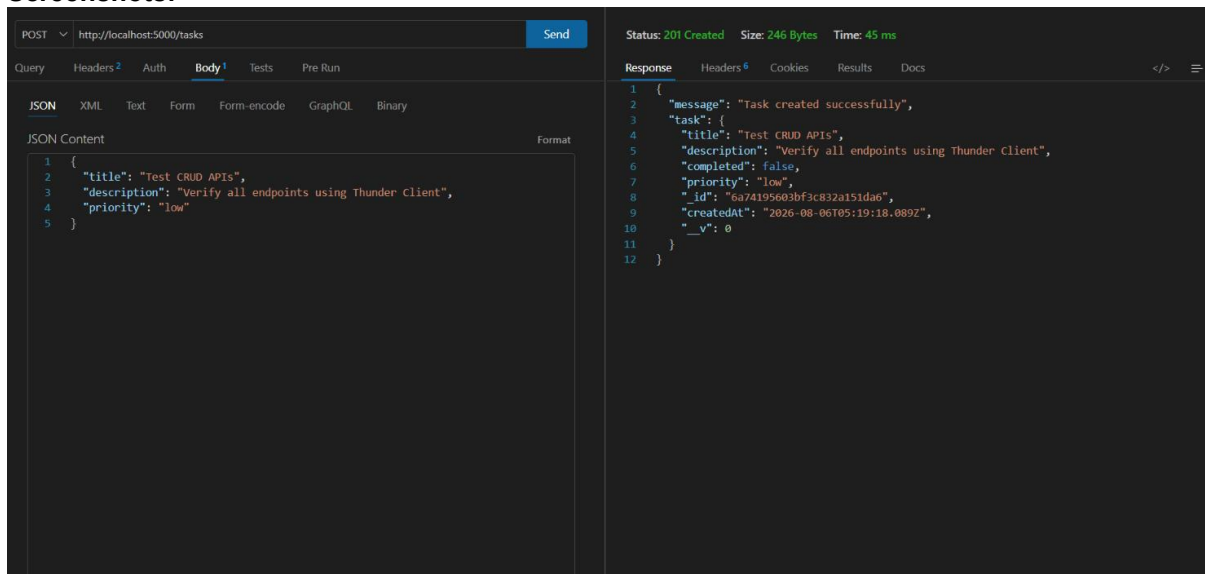


Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement:

To extend the Task Management REST API developed in Practical 4 by integrating MongoDB with Mongoose. The application connects to a MongoDB database, defines a validated Task schema, replaces the in-memory array with MongoDB CRUD operations, and performs Create, Read, Update, and Delete operations using Mongoose while ensuring proper schema validation and structured JSON error handling.

Screenshots:

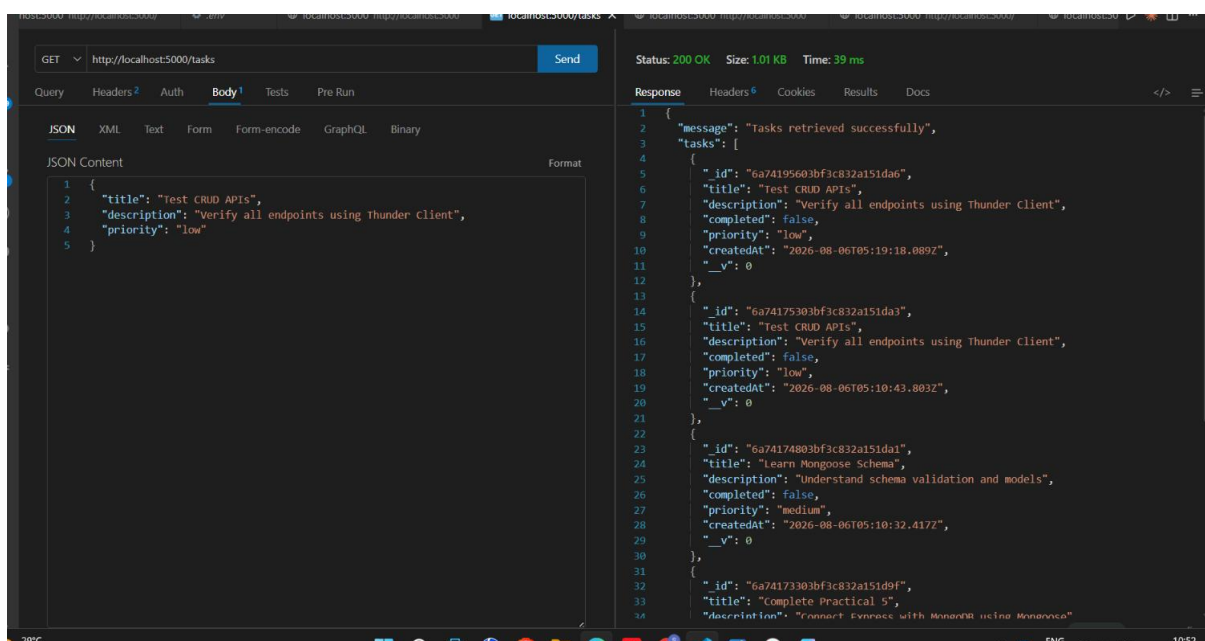


```

POST http://localhost:5000/tasks
{
  "title": "Test CRUD APIs",
  "description": "Verify all endpoints using Thunder Client",
  "priority": "low"
}

Status: 201 Created Size: 246 Bytes Time: 45 ms
{
  "message": "Task created successfully",
  "task": {
    "title": "Test CRUD APIs",
    "description": "Verify all endpoints using Thunder Client",
    "completed": false,
    "priority": "low",
    "_id": "6a74195603bf3c832a151da6",
    "createdAt": "2026-08-06T05:19:18.089Z",
    "__v": 0
  }
}

```



```

GET http://localhost:5000/tasks

Status: 200 OK Size: 1.01 KB Time: 39 ms
{
  "message": "Tasks retrieved successfully",
  "tasks": [
    {
      "_id": "6a74195603bf3c832a151da6",
      "title": "Test CRUD APIs",
      "description": "Verify all endpoints using Thunder Client",
      "completed": false,
      "priority": "low",
      "createdAt": "2026-08-06T05:19:18.089Z",
      "__v": 0
    },
    {
      "_id": "6a74175303bf3c832a151da3",
      "title": "Test CRUD APIs",
      "description": "Verify all endpoints using Thunder Client",
      "completed": false,
      "priority": "low",
      "createdAt": "2026-08-06T05:10:43.803Z",
      "__v": 0
    },
    {
      "_id": "6a74174803bf3c832a151da1",
      "title": "Learn Mongoose Schema",
      "description": "Understand schema validation and models",
      "completed": false,
      "priority": "medium",
      "createdAt": "2026-08-06T05:10:32.417Z",
      "__v": 0
    },
    {
      "_id": "6a74173303bf3c832a151d9f",
      "title": "Complete Practical 5",
      "description": "Connect Express with MongoDB using Mongoose"
    }
  ]
}

```

The screenshot displays two web browser windows. The top window is a REST client (likely Postman) showing a POST request to `http://localhost:5000/tasks`. The request body is a JSON object:

```
{  "title": "Complete Practical 5",  "description": "Connect Express with MongoDB",  "priority": "high"}
```

. The response status is 201 Created, with a size of 240 Bytes and a time of 37 ms. The response body is a JSON object:

```
{  "message": "Task created successfully",  "task": {    "title": "Complete Practical 5",    "description": "Connect Express with MongoDB",    "completed": false,    "priority": "high",    "_id": "6a741b3b03bf3c832a151da9",    "createdAt": "2026-08-06T05:27:23.107Z",    "__v": 0  }}
```

The bottom window is a MongoDB interface showing a collection named `tasks`. It displays a list of documents with the following fields: `completed`, `priority`, `createdAt`, and `__v`. The documents are:

- `{ "completed": false, "priority": "high", "createdAt": "2026-08-06T05:10:11.649+00:00", "__v": 0 }`
- `{ "_id": ObjectId('6a74174803bf3c832a151da1'), "title": "Learn Mongoose Schema", "description": "Understand schema validation and models", "completed": false, "priority": "medium", "createdAt": "2026-08-06T05:10:32.417+00:00", "__v": 0 }`
- `{ "_id": ObjectId('6a74175303bf3c832a151da3'), "title": "Test CRUD APIs", "description": "Verify all endpoints using Thunder Client", "completed": false, "priority": "low", "createdAt": "2026-08-06T05:10:43.803+00:00", "__v": 0 }`
- `{ "_id": ObjectId('6a74195603bf3c832a151da6'), "title": "Test CRUD APIs", "description": "Verify all endpoints using Thunder Client", "completed": false, "priority": "low", "createdAt": "2026-08-06T05:19:18.089+00:00", "__v": 0 }`

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tasks

localhost:27017 > AWDF_P5 > tasks

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN Reset Find Options

ADD DATA BULK EXPORT DATA EXPORT CODE

100 1 - 5 of 5

```
_id: ObjectId('6a7415f803bf3c832a151d9c')
title: "Study MongoDB"
description: "Practical 5"
completed: false
priority: "high"
createdAt: 2026-08-06T05:04:56.612+00:00
__v: 0
```

```
_id: ObjectId('6a74173303bf3c832a151d9f')
title: "Complete Practical 5"
description: "Connect Express with MongoDB using Mongoose"
completed: false
priority: "high"
createdAt: 2026-08-06T05:10:11.649+00:00
__v: 0
```

```
_id: ObjectId('6a74174803bf3c832a151da1')
title: "Learn Mongoose Schema"
description: "Understand schema validation and models"
completed: false
priority: "medium"
createdAt: 2026-08-06T05:10:32.417+00:00
__v: 0
```

```
_id: ObjectId('6a74175303bf3c832a151da3')
title: "Test CRUD APIs"
```

DELETE http://localhost:5000/tasks/6a74195603bf3c832a151da6 Send

Query Headers Auth Body Tests Pre Run

Query Parameters

parameter value

Status: 200 OK Size: 246 Bytes Time: 52 ms

Response Headers Cookies Results Docs

```
1 {
2   "message": "Task deleted successfully",
3   "task": {
4     "_id": "6a74195603bf3c832a151da6",
5     "title": "Test CRUD APIs",
6     "description": "Verify all endpoints using Thunder Client",
7     "completed": false,
8     "priority": "low",
9     "createdAt": "2026-08-06T05:19:18.089Z",
10    "__v": 0
11  }
12 }
```

PUT http://localhost:5000/tasks/6a7415f803bf3c832a151d9c Send

Query Headers Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
1 {
2   "title": "Study Mongoose",
3   "completed": true,
4   "priority": "medium"
5 }
```

Status: 200 OK Size: 218 Bytes Time: 41 ms

Response Headers Cookies Results Docs

```
1 {
2   "message": "Task updated successfully",
3   "task": {
4     "_id": "6a7415f803bf3c832a151d9c",
5     "title": "Study Mongoose",
6     "description": "Practical 5",
7     "completed": true,
8     "priority": "medium",
9     "createdAt": "2026-08-06T05:04:56.612Z",
10    "__v": 0
11  }
12 }
```

Student ID:24IT087

```

_id: ObjectId('6a7415f803bf3c832a151d9c')
title: "Study Mongoose"
description: "Practical 5"
completed: true
priority: "medium"
createdAt: 2026-08-06T05:04:56.612+00:00
__v: 0

_id: ObjectId('6a74173303bf3c832a151d9f')
title: "Complete Practical 5"
description: "Connect Express with MongoDB using Mongoose"
completed: false
priority: "high"
createdAt: 2026-08-06T05:10:11.649+00:00
__v: 0
```

Github Repo Link:

https://github.com/digishasavaliya/ITUE301_AWDF.git